Arithmetic Sequences and Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic sequences

Consider the sequence $7, 11, 15, 19, \ldots$

a State the common difference. $d = 4$

b Write a formula for the nth term. $a_n = 4n + 3$

c Find a_{20}. $a_{20} = 83$

(a) $d = 4$; (b) $a_n = 4n + 3$; (c) $a_{20} = 83$

2 [Foundational] • Target: Arithmetic sequences

In an arithmetic sequence, $a_3 = 10$ and $a_8 = 25$.

a Find the common difference and the first term. $5d = 15$ gives $d = 3$; $a_1 = 10 - 2(3) = 4$.

b Write a formula for a_n. $a_n = 4 + 3(n - 1) = 3n + 1$.

(a) $5d = 15$ gives $d = 3$; $a_1 = 10 - 2(3) = 4$.; (b) $a_n = 4 + 3(n - 1) = 3n + 1$.

3 [Foundational] • Target: Arithmetic series

Find the sum of the first 30 terms of the series $5 + 9 + 13 + \cdots$

$a_{30} = 5 + 29(4) = 121$, so $S_{30} = \frac{30(5 + 121)}{2} = 1890$.

4 [Foundational] • Target: Arithmetic series

Consider the multiples of 6 strictly between 100 and 500.

a How many are there? First 102, last 498: $n = \frac{498 - 102}{6} + 1 = 67$.

b Find their sum. $S = \frac{67(102 + 498)}{2} = 67 \times 300 = 20{,}100$.

(a) First 102, last 498: $n = \frac{498 - 102}{6} + 1 = 67$.; (b) $S = \frac{67(102 + 498)}{2} = 67 \times 300 = 20{,}100$.

5 [Foundational] • Target: Arithmetic series

A theatre has 18 seats in the first row, and each subsequent row has 3 more seats than the row before. How many seats are there in total in 25 rows?

$a_{25} = 18 + 24(3) = 90$, so $S_{25} = \frac{25(18 + 90)}{2} = 1350$ seats.

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6 [Foundational] • Target: Arithmetic means and sequence properties

Insert three arithmetic means between 5 and 25 (i.e., find b_1, b_2, b_3 so that $5, b_1, b_2, b_3, 25$ is arithmetic).

5 terms total, so $d = \frac{25-5}{4} = 5$. Means: $10, 15, 20$.

7 [Foundational] • Target: Arithmetic means and sequence properties

Given the arithmetic sequence with $a_1 = -8$ and $d = 3$, find the smallest n such that $a_n > 100$.

$a_n = -8 + 3(n-1) = 3n - 11 > 100$: $3n > 111$, $n > 37$: smallest integer $n = 38$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Arithmetic means and sequence properties

Prove that the sum of the first n positive odd integers $(1 + 3 + 5 + \cdots)$ equals n^2, using the arithmetic series formula.

The nth odd number is $a_n = 2n - 1$, with $a_1 = 1$, $d = 2$. $S_n = \frac{n(a_1 + a_n)}{2} = \frac{n(1 + 2n - 1)}{2} = \frac{n(2n)}{2} = n^2$.

9 [Intermediate] • Target: Applications

A person saves 100 riyals in week 1, and increases savings by 20 riyals each subsequent week. Find the total saved after 20 weeks.

$a_{20} = 100 + 19(20) = 480$. $S_{20} = \frac{20(100 + 480)}{2} = 5800$ riyals.

10 [Intermediate] • Target: Applications

A stack of logs has 20 logs in the bottom row, 19 in the next, decreasing by 1 each row, ending with 1 log in the top row. Find the total number of logs.

This is the sum $1 + 2 + \cdots + 20 = \frac{20(1 + 20)}{2} = 210$ logs.

11 [Intermediate] • Target: Visualizing an arithmetic sequence

The first 6 terms of the sequence $a_n = 3n - 1$ are plotted below. Use the graph to confirm the sequence is linear in n, and state the common difference from the graph.

The points $(1, 2), (2, 5), (3, 8), (4, 11), (5, 14), (6, 17)$ lie on a straight line, confirming the linear (arithmetic) pattern. The common difference $d = 3$ matches the constant rise between consecutive points.

12 [Intermediate] • Target: Arithmetic sequences in geometry

The interior angles of a pentagon form an arithmetic sequence. If the smallest angle is $80°$ and the angles sum to $540°$ (the pentagon angle sum), find the common difference.

$S_5 = \frac{5(2a_1 + 4d)}{2} = 540$:
$\frac{5(160 + 4d)}{2} = 540 \Rightarrow 5(160 + 4d) = 1080 \Rightarrow 160 + 4d = 216 \Rightarrow 4d = 56 \Rightarrow d = 14°$.

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13 [Intermediate] • Target: Finding a term given the sum

The sum of the first n terms of an arithmetic sequence with $a_1 = 3$ and $d = 4$ is $S_n = 210$. Find n.

$S_n = \frac{n(2(3)+(n-1)4)}{2} = \frac{n(4n+2)}{2} = n(2n+1) = 210 \Rightarrow 2n^2 + n - 210 = 0$. Quadratic formula: $n = \frac{-1 \pm \sqrt{1+1680}}{4} = \frac{-1 \pm 41}{4}$. Taking the positive root: $n = 10$.

14 [Intermediate] • Target: Arithmetic sequences with negative common difference

A sequence begins $50, 44, 38, 32, \ldots$ Find the first negative term.

$d = -6$: $a_n = 50 - 6(n-1) = 56 - 6n$. Need $56 - 6n < 0 \Rightarrow n > 9.33$: first negative term is $a_{10} = 56 - 60 = -4$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Proving an arithmetic sequence property • Cross-Lesson: 4.02 Ellipses

Prove that for any three consecutive terms a_{n-1}, a_n, a_{n+1} of an arithmetic sequence, the middle term equals the average of its neighbors: $a_n = \frac{a_{n-1} + a_{n+1}}{2}$.

By definition, $a_{n-1} = a_n - d$ and $a_{n+1} = a_n + d$. So $\frac{a_{n-1} + a_{n+1}}{2} = \frac{(a_n - d) + (a_n + d)}{2} = \frac{2a_n}{2} = a_n$, confirming the middle term is the average of its neighbors.

16 [Challenging] • Target: Arithmetic means and sequence properties • Cross-Lesson: 4.03 Hyperbolas

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n? Show your complete algebraic work.

Final Answer: $a_n = 5 + 3(n-1)$. The formula for the n-th term of an arithmetic sequence is given by $a_n = a_1 + d(n-1)$, where a_1 is the first term and d is the common difference. In this case, $a_1 = 5$ and $d = 3$. Substituting these values into the formula gives $a_n = 5 + 3(n-1)$. If we simplify this, we get $a_n = 5 + 3n - 3 = 3n + 2$. Therefore, the direct formula is $a_n = 5 + 3(n-1)$ or its simplified form $a_n = 3n + 2$. Choice A directly uses the definition. Choice B is the simplified form. Let'.

17 [Challenging] • Target: Applications • Cross-Lesson: 5.01 Vectors: Components and Operations

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

Final Answer: 5. The sum of the first n terms of a geometric sequence is given by the formula $S_n = a_1 \frac{r^n - 1}{r - 1}$, where a_1 is the first term and r is the common ratio. Given $a_1 = 5$ and $r = 2$, and $S_n = 315$, we can substitute these values into the formula: $315 = 5\frac{2^n - 1}{2 - 1}$ $315 = 5\frac{2^n - 1}{1}$ $315 = 5(2^n - 1)$ Divide both sides by 5: $\frac{315}{5} = 2^n - 1$ $63 = 2^n - 1$ Add 1 to both sides: $63 + 1 = 2^n$ $64 = 2^n$ Since $2^6 = 64$, we have $n = 6$. Let's ch.

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18 [Challenging] • Target: Visualizing an arithmetic sequence • Cross-Lesson: 5.02 The Dot Product

Consider an arithmetic sequence where the 5th term is 19 and the 11th term is 43. If the first term is denoted by a_1 and the common difference by d, find the value of $a_1 + d$.

Final Answer: 7. Let the arithmetic sequence be defined by $a_n = a_1 + (n-1)d$. We are given that $a_5 = 19$ and $a_{11} = 43$. From the given information, we can set up a system of two linear equations: 1. $a_1 + (5-1)d = 19 \Rightarrow a_1 + 4d = 19$ 2. $a_1 + (11-1)d = 43 \Rightarrow a_1 + 10d = 43$ Subtract equation (1) from equation (2): $(a_1 + 10d) - (a_1 + 4d) = 43 - 19$ $6d = 24$ $d = \frac{24}{6} = 4$ Now substitute the value of d back into equation (1): $a_1 + 4(4) = 19$ $a_1 + 16 = 19$.

19 [Challenging] • Target: Arithmetic sequences in geometry • Cross-Lesson: 5.03 Parametric Equations

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the 4th term of this sequence, a_4? Show your complete algebraic work.

Final Answer: 14. An arithmetic sequence is defined by the formula $a_n = a_1 + (n-1)d$, where a_n is the nth term, a_1 is the first term, and d is the common difference. In this problem, we are given $a_1 = 5$ and $d = 3$. We need to find the 4th term, so $n = 4$. Substituting these values into the formula: $a_4 = 5 + (4-1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$ Alternatively, we can list the terms: $a_1 = 5$ $a_2 = a_1 + d = 5 + 3 = 8$ $a_3 = a_2 + d = 8 + 3 = 11$.

20 [Challenging] • Target: Finding a term given the sum • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the fourth term of this sequence, a_4? Show your complete algebraic work.

Final Answer: 14. The formula for the n-th term of an arithmetic sequence is $a_n = a_1 + (n-1)d$. Given $a_1 = 5$ and $d = 3$, we want to find a_4. Substituting $n = 4$ into the formula: $a_4 = 5 + (4-1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$. Therefore, the fourth term of the sequence is 14. Common errors might occur from miscalculating $(n-1)$ or incorrectly applying the addition.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: General Term of an Arithmetic Sequence

State the formula for the n-th term a_n of an arithmetic sequence with first term a_1 and common difference d.

a. $a_n = a_1 + (n-1)d$ ✓

b. $a_n = a_1 \cdot d^{n-1}$

c. $a_n = a_1 + nd$

d. $a_n = \frac{a_1 + a_n}{2}$

In an arithmetic sequence, each term increases by d, so the n-th term is $a_n = a_1 + (n-1)d$.

22 [Foundational] • Target: General Term of an Arithmetic Sequence

Find the 20th term of the arithmetic sequence $4, 7, 10, 13, \ldots$

a. 61 ✓

b. 58

c. 64

d. 60

$a_1 = 4, d = 3$. $a_{20} = 4 + (20-1)(3) = 4 + 19(3) = 4 + 57 = 61$.

23 [Foundational] • Target: General Term of an Arithmetic Sequence

In an arithmetic sequence, $a_3 = 11$ and $a_8 = 31$. Find the first term a_1 and common difference d.

a. $a_1 = 3, d = 4$ ✓

b. $a_1 = 4, d = 3$

c. $a_1 = 2, d = 5$

d. $a_1 = 1, d = 4$

$a_8 - a_3 = 5d = 31 - 11 = 20 \Rightarrow d = 4$. $a_1 = a_3 - 2d = 11 - 2(4) = 3$.

24 [Foundational] • Target: Sum of an Arithmetic Series

State the two standard formulas for the sum S_n of the first n terms of an arithmetic series.

a. $S_n = \frac{n}{2}(a_1 + a_n)$ **and** $S_n = \frac{n}{2}[2a_1 + (n-1)d]$ ✓

b. $S_n = \frac{a_1(1 - r^n)}{1 - r}$

c. $S_n = n(a_1 + a_n)$

d. $S_n = \frac{a_1 + a_n}{2}$

Gauss's pairing formula gives $S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}[2a_1 + (n-1)d]$.

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25 [Foundational] • Target: Sum of an Arithmetic Series

Find the sum of the first 30 terms of the arithmetic series $2 + 5 + 8 + 11 + \ldots$

a. 1365 ✓ **b.** 1350

c. 2730 **d.** 1410

$n = 30, a_1 = 2, d = 3$. $S_{30} = \frac{30}{2}[2(2) + (29)(3)] = 15[4 + 87] = 15(91) = 1365$.

26 [Foundational] • Target: Sum of an Arithmetic Series

Find the sum of the positive integers from 1 to 100.

a. 5050 ✓ **b.** 5000

c. 5100 **d.** 10100

$S_{100} = \frac{100(101)}{2} = 50 \times 101 = 5050$.

27 [Foundational] • Target: Sigma Notation and Number of Terms

Evaluate $\sum_{k=1}^{25} (4k - 1)$.

a. 1275 ✓ **b.** 1300

c. 1250 **d.** 2550

First term $a_1 = 4(1) - 1 = 3$. Last term $a_{25} = 4(25) - 1 = 99$. Number of terms $n = 25$.
$S_{25} = \frac{25}{2}(3 + 99) = \frac{25}{2}(102) = 25(51) = 1275$.

28 [Foundational] • Target: Sigma Notation and Number of Terms

How many terms are in the finite arithmetic sequence $7, 11, 15, \ldots, 87$?

a. 21 ✓ **b.** 20

c. 22 **d.** 19

$a_n = a_1 + (n - 1)d \Rightarrow 87 = 7 + (n - 1)(4) \Rightarrow 80 = 4(n - 1) \Rightarrow 20 = n - 1 \Rightarrow n = 21$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Sigma Notation and Number of Terms

Evaluate $\sum_{k=5}^{20} 3k$.

a. 600 ✓

b. 570

c. 630

d. 300

Number of terms $n = 20 - 5 + 1 = 16$. $a_1 = 3(5) = 15$, $a_{16} = 3(20) = 60$.
$S = \frac{16}{2}(15 + 60) = 8(75) = 600$.

30 [Intermediate] • Target: Word Problems and Real-World Applications

An auditorium has 20 rows of seats. The first row has 15 seats, and each subsequent row has 2 more seats than the row before it. How many total seats are in the auditorium?

a. 680 ✓

b. 660

c. 700

d. 580

$a_1 = 15, d = 2, n = 20$. $S_{20} = \frac{20}{2}[2(15) + (19)(2)] = 10[30 + 38] = 10(68) = 680$.

31 [Intermediate] • Target: Word Problems and Real-World Applications

A runner trains by running 3 miles on day 1, and increases her daily distance by 0.5 miles each day. How many total miles will she have run in 14 days?

a. 87.5 miles ✓

b. 91 miles

c. 84 miles

d. 95.5 miles

$S_{14} = \frac{14}{2}[2(3) + (13)(0.5)] = 7[6 + 6.5] = 7(12.5) = 87.5$ miles.

32 [Intermediate] • Target: Word Problems and Real-World Applications

A clock chimes the number of hours on the hour (1 chime at 1:00, 2 chimes at 2:00, ..., 12 chimes at 12:00). How many total chimes does it make in a 12-hour period?

a. 78 ✓

b. 72

c. 84

d. 66

Sum $= \frac{12(13)}{2} = 6 \times 13 = 78$.

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33 [Intermediate] • Target: Word Problems and Real-World Applications

Find the 6th term of the arithmetic sequence with $a_1 = 2$ and common difference $d = -4$.

a. **-18** ✓

b. -14

c. -26

d. -22

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = 2 + (6-1)(-4) = -18$. Therefore, the verified correct option is -18.

34 [Intermediate] • Target: Arithmetic sequences

Find the 8th term of the arithmetic sequence with $a_1 = 0$ and common difference $d = -4$.

a. -24

b. **-28** ✓

c. -36

d. -32

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = 0 + (8-1)(-4) = -28$. Therefore, the verified correct option is -28.

35 [Intermediate] • Target: Arithmetic series

Find the 6th term of the arithmetic sequence with $a_1 = 6$ and common difference $d = -5$.

a. -29

b. **-19** ✓

c. -14

d. -24

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = 6 + (6-1)(-5) = -19$. Therefore, the verified correct option is -19.

36 [Intermediate] • Target: Arithmetic means and sequence properties

Find the 5th term of the arithmetic sequence with $a_1 = 3$ and common difference $d = -5$.

a. **-17** ✓

b. -22

c. -12

d. -27

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = 3 + (5-1)(-5) = -17$. Therefore, the verified correct option is -17.

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37 [Intermediate] • Target: Applications

Find the 5th term of the arithmetic sequence with $a_1 = 8$ and common difference $d = -5$.

a. -17

b. -22

c. -7

d. **-12 ✓**

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = 8 + (5-1)(-5) = -12$. Therefore, the verified correct option is -12.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Visualizing an arithmetic sequence • Cross-Lesson: 5.02 The Dot Product

Find the 6th term of the arithmetic sequence with $a_1 = 0$ and common difference $d = -6$.

a. **-30 ✓**

b. -36

c. -24

d. -42

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = 0 + (6-1)(-6) = -30$. Therefore, the verified correct option is -30.

39 [Challenging] • Target: Arithmetic sequences in geometry • Cross-Lesson: 5.03 Parametric Equations

Find the 4th term of the arithmetic sequence with $a_1 = 5$ and common difference $d = -5$.

a. -15

b. **-10 ✓**

c. -5

d. -20

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = 5 + (4-1)(-5) = -10$. Therefore, the verified correct option is -10.

40 [Challenging] • Target: Finding a term given the sum • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the 8th term of the arithmetic sequence with $a_1 = 3$ and common difference $d = -2$.

a. -13

b. -15

c. **-11 ✓**

d. -9

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = 3 + (8-1)(-2) = -11$. Therefore, the verified correct option is -11.

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41 [Challenging] • Target: Arithmetic sequences with negative common difference • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the 6th term of the arithmetic sequence with $a_1 = 1$ and common difference $d = -5$.

a. -29

b. -19

c. -34

d. **-24** ✓

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = 1 + (6-1)(-5) = -24$. Therefore, the verified correct option is -24.

42 [Challenging] • Target: Proving an arithmetic sequence property • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Find the 6th term of the arithmetic sequence with $a_1 = -2$ and common difference $d = -1$.

a. **-7** ✓

b. -8

c. -6

d. -9

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = -2 + (6-1)(-1) = -7$. Therefore, the verified correct option is -7.

43 [Challenging] • Target: General Term of an Arithmetic Sequence • Cross-Lesson: 7.02 Logarithms and Their Properties

Find the 4th term of the arithmetic sequence with $a_1 = 6$ and common difference $d = -5$.

a. -4

b. -19

c. -14

d. **-9** ✓

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = 6 + (4-1)(-5) = -9$. Therefore, the verified correct option is -9.

44 [Challenging] • Target: Sum of an Arithmetic Series • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Find the 8th term of the arithmetic sequence with $a_1 = -2$ and common difference $d = 6$.

a. **40** ✓

b. 46

c. 34

d. 52

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = -2 + (8-1)(6) = 40$. Therefore, the verified correct option is 40.

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45 [Challenging] • Target: Sigma Notation and Number of Terms • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the 5th term of the arithmetic sequence with $a_1 = 6$ and common difference $d = -2$.

a. -4

b. **-2** ✓

c. 0

d. -6

Applying Sequences & Series principles, $a_n = a_1 + (n-1)d = 6 + (5-1)(-2) = -2$. Therefore, the verified correct option is -2.

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